

Решение задания:

- Создаем директорию /Docker:

```
sudo mkdir Docker
```

```
cd Docker
```

- в ней директорию для выполнения задания:

```
sudo mkdir ServerISQL
```

```
cd ServerISQL
```

- в ней пустые директории /src – для файла index.php , /dbfile – для файлов БД, /dbDockerfile – для Dockerfile контейнера с БД:

```
sudo mkdir src
```

```
sudo mkdir dbfile
```

```
sudo mkdir dbDockerfile
```

- создаем файл Dockerfile для сервера:

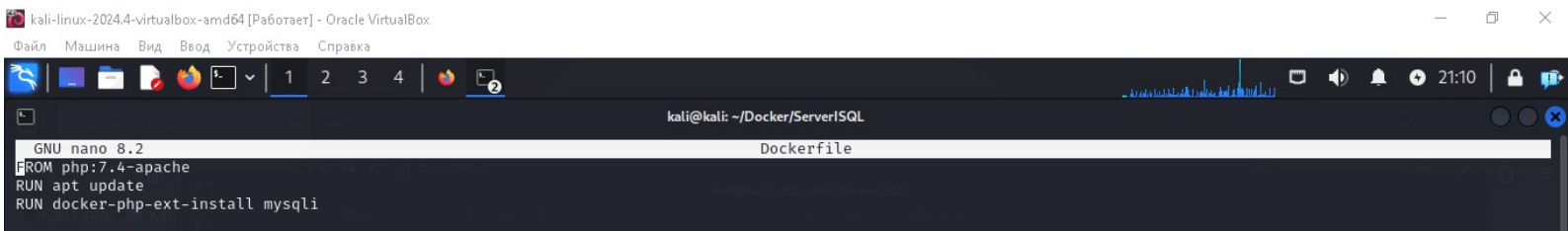
```
nano Dockerfile
```

- с содержанием:

```
FROM php:7.4-apache
```

```
RUN apt update
```

```
RUN docker-php-ext-install mysqli
```



The screenshot shows a terminal window titled "kali@kali: ~/Docker/ServerISQL". The terminal displays the content of the Dockerfile, which is being edited with nano 8.2. The content is as follows:

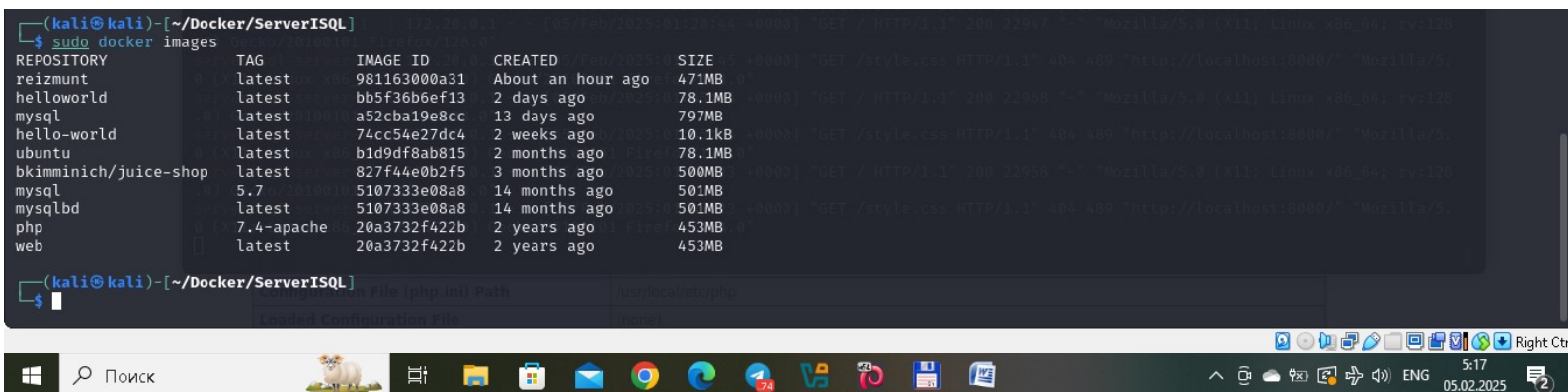
```
FROM php:7.4-apache
RUN apt update
RUN docker-php-ext-install mysqli
```

- собираем контейнер сервера с названием по фамилии:

```
sudo docker build -t reizmunt .
```

- убеждаемся в наличии образа контейнера:

```
sudo docker images
```



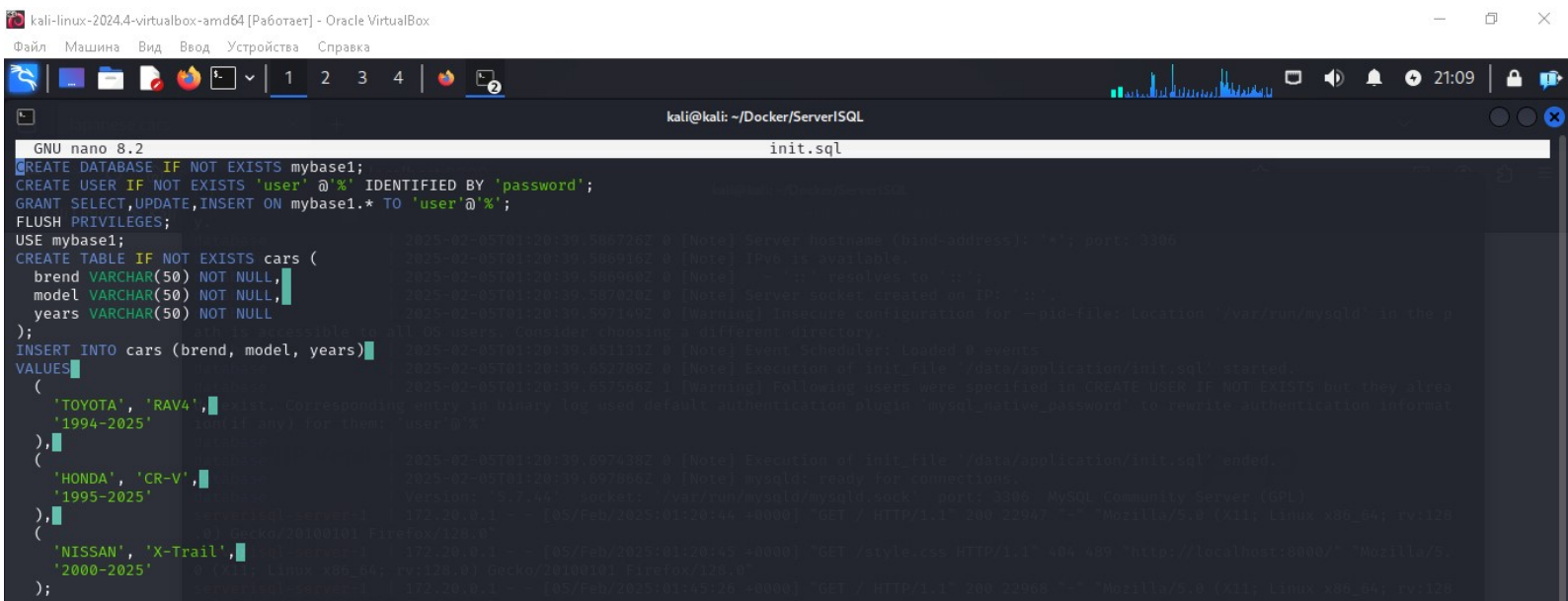
The screenshot shows a terminal window titled "(kali@kali) - [~/Docker/ServerISQL]". The terminal displays the output of the command "sudo docker images". The output is a table with columns: REPOSITORY, TAG, IMAGE ID, CREATED, and SIZE. The table lists several images, including reizmunt, hello-world, mysql, and php.

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
reizmunt	latest	981163000a31	About an hour ago	471MB
hello-world	latest	bb5f36b6ef13	2 days ago	78.1MB
mysql	latest	a52cba19e8cc	13 days ago	797MB
hello-world	latest	74cc54e27dc4	2 weeks ago	10.1kB
ubuntu	latest	b1d9df8ab815	2 months ago	78.1MB
bkimminich/juice-shop	latest	827f44e0b2f5	3 months ago	500MB
mysql	5.7	5107333e08a8	14 months ago	501MB
mysqlbd	latest	5107333e08a8	14 months ago	501MB
php	7.4-apache	20a3732f422b	2 years ago	453MB
web	latest	20a3732f422b	2 years ago	453MB

- здесь же создаем файл init.sql:

```
nano init.sql
```

- с содержанием:



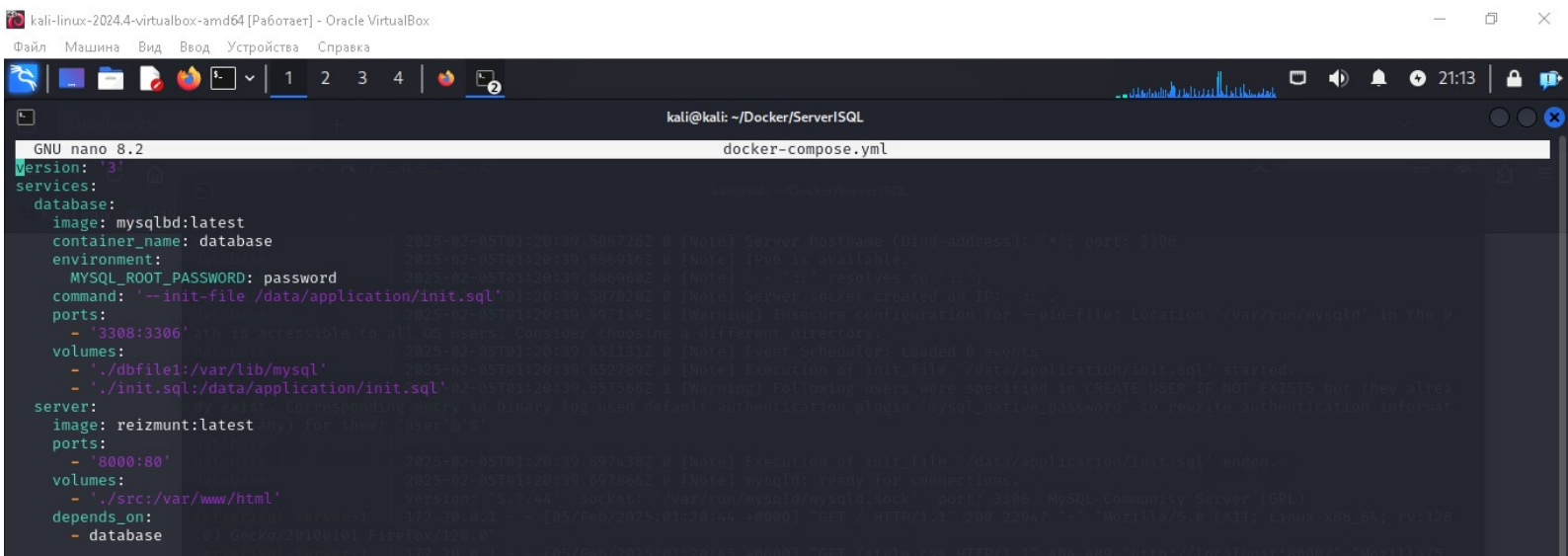
```
kali@kali: ~/Docker/ServerISQL
GNU nano 8.2                                init.sql
CREATE DATABASE IF NOT EXISTS mybase1;
CREATE USER IF NOT EXISTS 'user' @'%' IDENTIFIED BY 'password';
GRANT SELECT,UPDATE,INSERT ON mybase1.* TO 'user'@'%';
FLUSH PRIVILEGES;
USE mybase1;
CREATE TABLE IF NOT EXISTS cars (
  brend VARCHAR(50) NOT NULL,
  model VARCHAR(50) NOT NULL,
  years VARCHAR(50) NOT NULL
);
INSERT INTO cars (brend, model, years)
VALUES
  ('TOYOTA', 'RAV4',
  '1994-2025'
),
  ('HONDA', 'CR-V',
  '1995-2025'
),
  ('NISSAN', 'X-Trail',
  '2000-2025'
);
```

- здесь мы создаем БД mybase1 с логином “user”, паролем “password”, таблицей “cars” со столбцами “brend”, “model”, “years” и добавляем в нее три строки: 'TOYOTA', 'RAV4', '1994-2025', 'HONDA', 'CR-V', '1995-2025', 'NISSAN', 'X-Trail', '2000-2025'.

- создаем файл docker-compose.yml:

nano docker-compose.yml

- с содержанием:



```
kali@kali: ~/Docker/ServerISQL
GNU nano 8.2                                docker-compose.yml
version: '3'
services:
  database:
    image: mysqlbd:latest
    container_name: database
    environment:
      MYSQL_ROOT_PASSWORD: password
    command: '--init-file /data/application/init.sql'
    ports:
      - '3308:3306'
    volumes:
      - './dbfile1:/var/lib/mysql'
      - './init.sql:/data/application/init.sql'
  server:
    image: reizmunt:latest
    ports:
      - '8000:80'
    volumes:
      - './src:/var/www/html'
    depends_on:
      - database
```

- переходим в директорию dbDockerfile и в нем создаем файл Dockerfile для БД:

cd dbDockerfile

nano Dockerfile

- с содержанием:

FROM mysql:5.7

- и собираем образ БД:

sudo docker build -t mysqlbd

- возвращаемся в директорию ServerISQL , переходим в директорию /src и создаем в нем файл index.php:

cd -

nano index.php

- с содержанием:

```

kali-linux-2024.4-virtualbox-en64 [Работает] - Oracle VirtualBox
Файл Машина Вид Ввод Устройства Справка

kali@kali: ~/Docker/ServerMySQL/src
GNU nano 8.2 index.php
<html lang="en">
<head>
  <title>Japanese cars</title>
  <link rel="stylesheet" href="style.css" type="text/css" />
</head>
<body class=center>
  <h1>Japanese cars</h1>
  <table>
    <tr>
      <th>Brend</th>
      <th>Model</th>
      <th>Years</th>
    </tr>
    <?php
      $mysql = new mysqli("database", "user", "password", "mybase1");
      $result = $mysql->query("SELECT * FROM cars");
      foreach ($result as $row) {
        echo "
        <tr>
          <td>{$row['brend']}</td>
          <td>{$row['model']}</td>
          <td>{$row['years']}</td>
        </tr>
        "; } ?>
      </table>
    <?php
      phpinfo();
  </body>
</html>

```

2025-02-05T01:20:39.000720Z [Note] Server hostname (bind-address): '*'; port: 3306
 2025-02-05T01:20:39.000910Z [Note] IPv6 is available.
 2025-02-05T01:20:39.001002Z [Note] --user='root' resolves to id 1
 2025-02-05T01:20:39.001282Z [Note] Server socket created on IP: '*'
 2025-02-05T01:20:39.001492Z [Warning] Insecure configuration for --pid-file: location '/var/run/mysqld' in the p
 is accessible to all users, consider choosing a different directory.
 2025-02-05T01:20:39.051111Z [Note] Event Scheduler: Loaded 0 events
 2025-02-05T01:20:39.051190Z [Note] mysqld: ready for connections.
 2025-02-05T01:20:39.051506Z [Warning] Following users were specified in CREATE USER IF NOT EXISTS but they already exist: corresponding entry for user 'root'@'localhost' in 'mysql_native_password' table; authentication informat

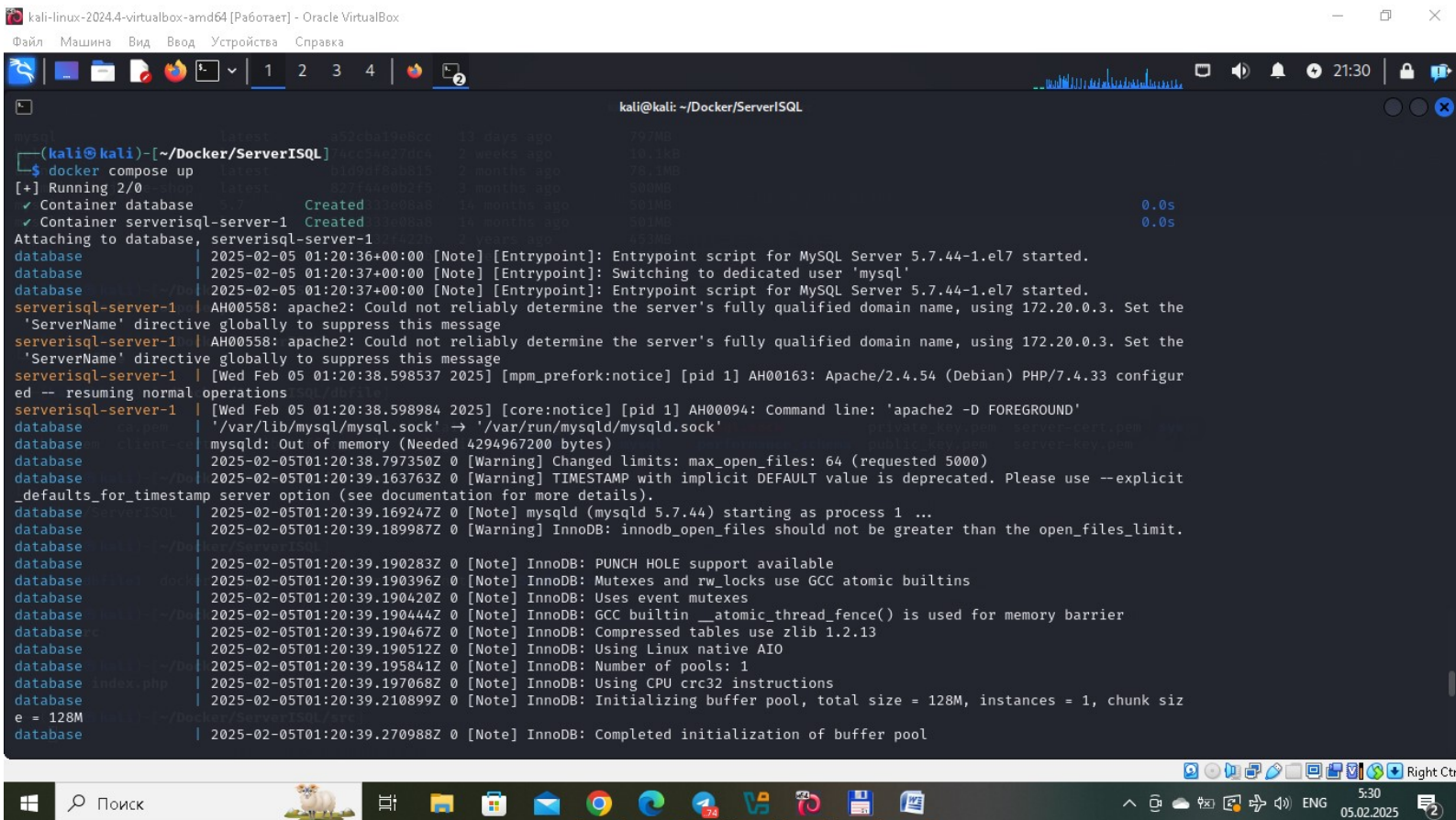
```

mysql_install_db --user=root
2025-02-05T01:20:39.057182Z [Note] Execution of init file '/data/application/init.sql' started.
2025-02-05T01:20:39.057662Z [Warning] Following users were specified in CREATE USER IF NOT EXISTS but they already exist: corresponding entry for user 'root'@'localhost' in 'mysql_native_password' table; authentication informat
2025-02-05T01:20:39.057662Z [Note] mysqld: ready for connections.
Version: 5.7.44  socket: /var/run/mysqld/mysqld.sock  port: 3306  MySQL Community Server (GPL)
172.20.0.1 -- (105/Feb/2025:01:20:44 +0000) "GET / HTTP/1.1" 200 22907 "-" Mozilla/5.0 (X11; Linux x86_64; rv:128
172.20.0.1 -- (105/Feb/2025:01:20:45 +0000) "GET /style.css HTTP/1.1" 404 409 "http://localhost:8080/" Mozilla/5.
172.20.0.1 -- (105/Feb/2025:01:20:45 +0000) "GET / HTTP/1.1" 200 22908 "-" Mozilla/5.0 (X11; Linux x86_64; rv:128
172.20.0.1 -- (105/Feb/2025:01:20:46 +0000) "GET / HTTP/1.1" 200 22908 "-" Mozilla/5.0 (X11; Linux x86_64; rv:128
172.20.0.1 -- (105/Feb/2025:01:20:46 +0000) "GET /style.css HTTP/1.1" 404 409 "http://localhost:8080/" Mozilla/5.
172.20.0.1 -- (105/Feb/2025:01:20:47 +0000) "GET / HTTP/1.1" 200 22958 "-" Mozilla/5.0 (X11; Linux x86_64; rv:128
172.20.0.1 -- (105/Feb/2025:01:20:47 +0000) "GET /style.css HTTP/1.1" 404 409 "http://localhost:8080/" Mozilla/5.

```

- возвращаемся в предыдущую директорию, в которой находится файл `docker-compose.yml` и запускаем `docker compose`:

```
sudo docker compose up
```



- открываем браузер, набираем в адресной строке **localhost:8000** и видим отображение таблицы и phpinfo:

